Adult Tachycardia with a Pulse

Appropriateness for Clinical Condition
Tachycardias requiring intervention are typically >150bpm

Identify and Treat Underlying Cause
- Maintain patient airway, assist breathing as necessary
- Titrate oxygen to maintain Sp02 >94%
- Routine vital sign assessment
- 12-lead ECG (do not delay treatment for severe cases)
- Consider Tachycardia due to shock (e.g. sepsis)

Consider ALS Back-Up

Persistent Tachycardia Causing:
- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic Chest Discomfort?
- Acute Heart Failure?

YES

Consider Adenosine
If regular
1st Dose: 6mg IV
2nd Dose: 12mg IV
P

Synchronized Cardioversion
Consider Sedation
AP

NO

Narrow Complex Tachycardia QRS < 0.12s

Vagal Maneuvers — EMT

Consider IV Fluid Bolus
250ml IV
EMT

Adenosine
If regular
1st Dose: 6mg IV
2nd Dose: 12mg IV
P

Wide Complex Tachycardia QRS > 0.12s

Consider IV Fluid Bolus
250ml IV
EMT

Consider Adenosine
If regular
1st Dose: 6mg IV
2nd Dose: 12mg IV
P

Amiodarone Infusion
150mg /10mins IV
- Repeat dose if tachycardia persist or recurs.
- Follow with 1mg/min infusion if rhythm reverts.

AP

If Refractory
- Consider underlying causes
- Consider telemedicine consultation

Patient Care Goals

1. Identify patients with tachycardia requiring intervention.
2. Identification of underlying causes.
3. Restoration normal rhythm and hemodynamic stability.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult patients presenting with tachycardia and a palpable pulse (typically >150 bpm).
- Conscious and unconscious adult patients

Exclusion Criteria

- Tachycardia with a suspected non-cardiac cause which may require other treatment (e.g. sepsis or other physiological causes).
- Pediatric patients <13 years.

Key Patient Assessment Points

1. **12-Lead ECG**
2. **Identification of cardiac rhythm**
    - Such as: VT, SVT, Sinus Tachycardia, other.
3. **Identification and Treatment of other causes:**
    - Consider toxins/drugs.
    - Physiological Causes
        - Sepsis
        - Pain/anxiety
        - Hypovolaemia
        - Anaemia
        - Street drugs
        - Hyperthyroidism
4. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
    - Red Priority:
        - Ventricular Tachycardia (VT)
        - Supraventricular Tachycardia (SVT)
        - Dysrhythmia with Signs of Severe Shock or hypotension

Patient Management & Interventions

1. **Perform Primary Survey**
    - As per Universal Care CPG.
2. **Attach Cardiac Monitor**
    - Assess Vital Signs
    - Assess underlying cardiac rhythm
    - **Perform 12-Lead ECG**
        - Performing a 12-Lead ECG may not be an immediate priority in unconscious patients or patients who have significant issues with their primary survey, or have a life-threatening arrhythmia.
3. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
4. **Identify and treat underlying causes**
5. **Establish IV Access**
    - Consider IO access if gaining IV access may be delayed or is otherwise not feasible.
6. **Assess for Signs of Instability**
    - Hypotension
    - Acutely altered mental state

- Signs of shock
- Ischaemic chest discomfort
- Acute heart failure

7. **If Patient is Hemodynamically Unstable**
   - **Request for ALS back-up**
   - **Consider Adenosine**
     - 1st Dose: 6mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
     - 2nd Dose: 12mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
     - Indicated for:
       - Regular tachycardias. And/
       - Patients who are not significantly compromised, or/
       - Where sedation is not available and cardioversion without sedation would be inappropriate.
   - **Perform Synchronized Cardioversion**
     - Consider sedation
     - Select Joule settings appropriate to the underlying rhythm. (see below).
     - If rhythm does not revert:
       - Increase Joule settings (see below) and perform synchronized cardioversion.
   - If the patient deteriorates into cardiac arrest, go to Cardiac Arrest CPG.
8. **If Patient is Hemodynamically Stable**
   - **Narrow Complex Tachycardia** (HR > 150/min with QRS < 0.12 sec)
     - **Vagal Stimulation** (If REGULAR)
       - Preferred Method: Modified Valsalva.
       - Alternative Methods: Valsalva, ice water applied to face, coughing, breath-holding.
     - **IV Fluid Bolus**
       - 250ml IV/IO crystalloid bolus.
       - Reassess frequently for fluid overload. Stop administration if fluid overload presents.
     - **Adenosine** (if REGULAR)
       - 1st Dose: 6mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
       - 2nd Dose: 12mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
   - **Wide Complex Tachycardia** (HR > 150/min with QRS > 0.12 sec)
     - **Adenosine** (if REGULAR)
       - 1st Dose: 6mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
       - 2nd Dose: 12mg IV/IO, rapid injection, with 20ml Normal Saline Flush.
     - **IV Fluid Bolus**
       - 250ml IV/IO crystalloid bolus.
       - Reassess frequently for fluid overload. Stop administration if fluid overload presents.
     - **Amiodarone**
       - Amiodarone Infusion:
         - 150mg infused over 10 minutes IV/IO. (15mg/min)
         - Diluted in 5% dextrose in 100ml (D5W).
       - Repeat dose (150mg) if rhythm fails to revert.
       - Follow with a 1 mg/min infusion.
9. **If Torsades de Pointes**
   - Consider Asynchronous Defibrillation
   - Administer Magnesium 2g IV over 10 min.
10. **If Rhythm is Refractory** to interventions:
    - Consider underlying causes.
    - Consider Tele-EMS consultation.
11. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Ensure equipment for cardiac arrest management is readily available during cardioversion.
- Monitor for signs of fluid overload, especially with IV fluid administration.
- Avoid vagal stimulation if varying R-R intervals or atrial fibrillation is present.

Clinical Insights & Additional Information

- **Preparing for Cardiac Arrest**
  - Electrical and chemical cardioversion have a risk of patient deterioration into cardiac arrest. When cardioverting any patient, either with electricity or medication, equipment to manage cardiac arrest should be readily available.
- **Compromised / Unstable Patients**
  - Synchronized Cardioversion is generally the first-line treatment.
- **Joule Settings for Synchronized Cardioversion**

	Rhythm	Initial Dose	Subsequent Dose
LIFEPAK 15	Wide Complex Tachycardia	100 J	200 J, 300 J, 360 J
	Supraventricular Tachycardia (SVT)	50 J	100 J, 200 J, 300 J, 360 J
	Atrial Fibrillation (AF)	120 J	200 J, 300 J, 360 J
ZOLL X-SERIES	Wide Complex Tachycardia	100 J	120 J, 150 J, 200 J
	Supraventricular Tachycardia (SVT)	50 J	120 J, 150 J, 200 J
	Atrial Fibrillation (AF)	120 J	150 J, 180 J, 200 J

Documentation

- Identification: Document patient demographics and clinical condition.
- Mechanism: Note suspected cause of tachycardia (e.g., shock, sepsis, hyperthyroidism).
- Injuries/Illness: Record signs of instability (e.g., hypotension, altered mental state).
- Signs/Symptoms: Include vital signs, ECG findings, and response to interventions.
- Treatment: Detail medications (e.g., adenosine doses, amiodarone infusion), cardioversion attempts, and vagal manoeuvres.
- Allergies: Confirm and document any known allergies.
- Medications: List patient's current medications.
- Background: Provide medical history relevant to tachycardia.
- Other: Note ALS back-up request, telemedicine consultation, and specialist advice sought.

Key Performance Indicators

1. Perform synchronized cardioversion on all hemodynamically compromised patients.

References

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